

The record costs recoil: a floor for every probe state

For every protocol of momentum-transfer marks with their pointers read, with probes in any states whatever (Gaussian or not, squeezed, correlated, mixed, grid states), deciding between free motion and a constant force F over a duration τ at error probability ϵ requires

$$s \sum_j \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon), \quad s = \frac{F\tau^2}{2m},$$

where s is Newton's sagitta for the interval and Δ_j is the spread of the impulse delivered by mark j . The resolutions of the marks do not enter. For a general force the sagitta is replaced by the uniform distance of the forced displacement from straight motion (Theorem R). At certain decision the bound is $s \sum_j \Delta_j \geq 4\pi\hbar = 2h$. The constant $\arcsin(1 - 2\epsilon)$ comes from running the proof with the Bures angle in place of total variation (Theorem P of the path-length note, 2026-09-23); the proof as written below gives the weaker $1 - 2\epsilon$.

The theorem answers the question the non-Gaussian probes raise. The floor $\tau\Delta E \geq 24z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$ of the mark-cost note is a theorem about Gaussian probes: a grid state with zero position-momentum correlation decides the comparison at any force (Proposition G), so no bound on second moments alone survives. What does survive, for every probe, is the recoil. A pointer registers a displacement only through the spread of its conjugate momentum, and that momentum is the impulse the body receives. This is M3 in its universal form: a record of the sagitta must leave the delivered impulse undetermined by at least $8\hbar \arcsin(1 - 2\epsilon)/s$ in total. The three-mark protocol of Proposition D, with the best Gaussian probe, comes within a factor $z_{1-\epsilon}/\arcsin(1 - 2\epsilon)$ of the bound at every recoil, about 1.47 at five per cent error (Section 4). Exploratory; no ledger promotion.

1. Setting

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, mass m , on $[0, \tau]$. Marks $j = 1, \dots, k$ act at times t_j : the body is coupled to probe j , prepared in a state φ_j (pure or mixed, arbitrary), by $U_j = \exp(-\frac{i}{\hbar}\lambda_j\hat{y}\hat{P}_j)$, and the pointer $\hat{Q}_j$ is read afterwards with no further evolution. Under hypothesis I the body moves freely; under F it feels a force whose displacement relative to free motion from rest at the origin is $P(t)$, with $P(0) = P'(0) = 0$; for the constant force $P(t) = Ft^2/2m$. The body's initial state is unknown: a test **decides at error** ϵ when its equal-prior error probability is at most ϵ for every initial state under I and every initial state under F. Write $\hat{N}_j = \hat{Q}_j/\lambda_j$ and $\hat{D}_j = -\lambda_j\hat{P}_j$ for the error and the delivered impulse of mark j , as in Theorem A, and $\Delta_j = \lambda_j\Delta_{\varphi_j}\hat{P}_j$ for its recoil spread.

Lemma 1 (the record). In the Heisenberg picture the j -th reading is

$$\hat{R}_j = \hat{y} + \frac{\hat{p}t_j}{m} + \theta P(t_j) + \sum_{i: t_i < t_j} \frac{t_j - t_i}{m} \hat{D}_i + \hat{N}_j,$$

with $\theta = 0$ under I and $\theta = 1$ under F, and the initial pointer $\hat{Q}_j$ appears in $\hat{R}_j$ alone.

Proof. Theorem A of the mark-cost note for each coupling, the linear free evolution between marks, and the c-number force, which adds $P(t)$ to $\hat{y}(t)$. The pointer $\hat{Q}_j$ is shifted by the coupling and coupled to nothing afterwards. $\square$

Lemma 2 (the signal moves into the pointers). For real a, b let the body's initial state under F be the state under I translated by $-a$ in position and $-mb$ in momentum. Then the record under F has the distribution of the record under I with each probe state φ_j replaced by its translate by $\lambda_j c_j$ in $\hat{Q}_j$, where $c_j = P(t_j) - a - bt_j$.

Proof. The translation of the body replaces $\hat{y}, \hat{p}$ in Lemma 1 by $\hat{y} - a, \hat{p} - mb$, which shifts $\hat{R}_j$ by $-a - bt_j$; the force adds $P(t_j)$. The net effect is the c-number shift $\hat{R}_j \mapsto \hat{R}_j + c_j$. Translating

probe j by $\lambda_j c_j$ in $\hat{Q}_j$ shifts $\hat{N}_j$ by c_j and leaves every $\hat{D}_i$ and every other reading unchanged, by the last clause of Lemma 1. The record operators commute, so their joint distribution is fixed by these shifts and the states. $\square$

2. The theorem

Theorem R. Every protocol deciding at error ϵ satisfies

$$1 - 2\epsilon \leq \frac{1}{\hbar} \min_{a,b} \sum_j \Delta_j |P(t_j) - a - bt_j|.$$

Proof. Fix a, b and any initial state σ under I, and take under F the translate of Lemma 2. A test with error at most ϵ against this pair has $\text{TV} \geq 1 - 2\epsilon$ between the two record distributions. By Lemma 2 those are the records of one fixed protocol, with body state σ , applied to the probe states $\otimes_j \varphi_j$ and $\otimes_j \varphi'_j$, φ'_j the translate by $q_j = \lambda_j c_j$. The data-processing inequality and the triangle inequality over the factors give

$$\text{TV} \leq \frac{1}{2} \|\otimes_j \varphi_j - \otimes_j \varphi'_j\|_1 \leq \sum_j \frac{1}{2} \|\varphi_j - \varphi'_j\|_1.$$

The translate is $e^{-iq_j \hat{P}_j / \hbar} \varphi_j e^{iq_j \hat{P}_j / \hbar}$. By Mandelstam–Tamm for a pure state, and through a purification for a mixed one exactly as in Theorem 2 of the probabilistic note, $\frac{1}{2} \|\varphi_j - \varphi'_j\|_1 \leq |q_j| \Delta_{\varphi_j} \hat{P}_j / \hbar = \Delta_j |c_j| / \hbar$. Minimize over a, b . $\square$

Corollary R1 (the Galileo comparison). For $P(t) = Ft^2/2m$, the line $a + bt$ that best approximates P uniformly on $[0, \tau]$ is $\frac{F}{2m}(\tau t - \tau^2/8)$, and $|P(t) - a - bt| \leq F\tau^2/16m = s/8$ there, with equality at $t = 0, \tau/2, \tau$. Hence

$$s \sum_j \Delta_j \geq 8(1 - 2\epsilon)\hbar.$$

For a general force the same step gives $E_1(P) \sum_j \Delta_j \geq (1 - 2\epsilon)\hbar$, with $E_1(P) = \min_{a,b} \max_{[0,\tau]} |P(t) - a - bt|$ the uniform distance of the forced displacement from straight motion. $\square$

The mechanism is the conjugacy of Theorem A read in the other direction. The record of the force is a translation of the pointers, and a pointer feels a translation only through the spread of the operator that generates it, which is its momentum $\hat{P}_j$. That same momentum, times λ_j , is the impulse the body receives. So sensitivity is paid in recoil at the exchange rate $\hbar$, whatever the pointer's state, and a resolution bought by squeezing, correlation or a grid is bought with recoil.

For adaptive protocols, in which couplings and times depend on earlier readings, the hybrid argument of the probabilistic note gives the same bound with each $\Delta_j |c_j|$ replaced by its supremum over the choices available at step j . The proof is written for product probe states; for probes correlated across marks the translations combine into one unitary generated by $\sum_j q_j \hat{P}_j$, the $\hat{P}_j$ commute, and the seminorm property $\Delta(\sum_j q_j \hat{P}_j) \leq \sum_j |q_j| \Delta \hat{P}_j$ gives the same bound.

Prior art. The step that bounds a pointer's response to a translation by its momentum spread is the standard displacement-sensing bound, the quantum Fisher information of a translation being $4\Delta \hat{P}^2 / \hbar^2$. What is specific here is the identification of the pointer momentum with the recoil delivered to the body, and the affine minimisation that brings in the sagitta. The standard quantum limit (Caves et al. 1980; Braginsky and Khalili 1992) bounds the product of imprecision and back-action; Theorem R bounds the back-action alone against the signal, whatever the imprecision.

3. The Gaussian floor needs Gaussian probes

Proposition G. In the three-mark protocol of Proposition D of the mark-cost note (marks at $0, \tau/2, \tau$, test $R_1 - 2R_2 + R_3$, outer marks sharp), let the middle probe run through a family of approximate grid states with real wavefunctions whose two stabilizer expectations tend to 1, such as Gaussian teeth of width w under a Gaussian envelope of width of order $1/w$, as $w \rightarrow 0$. Each has $\text{Cov}(\hat{Q}_2, \hat{P}_2) = 0$, so its correlation ρ vanishes. For every F whose signal $d = u^\top P = F\tau^2/4m$ is not a multiple of $\sigma = \sqrt{2\pi\hbar|u_2 S_2|/m} = \sqrt{2\pi\hbar\tau/m}$, the error probability tends to zero along the family, while the recoil Δ_2 diverges, as Theorem R requires.

Proof. The statistic's noise is $-2\hat{N}_2 + \frac{\tau}{2m}\hat{D}_2$ plus the vanishing contributions of the outer marks, that is $-(\alpha\hat{Q}_2 + \beta\hat{P}_2)$ with $\alpha = 2/\lambda_2$ and $\beta = \tau\lambda_2/2m$, so $\alpha\beta = |u_2 S_2|/m = \tau/m$ independent of λ_2 . Write $\hat{S} = \alpha\hat{Q}_2 + \beta\hat{P}_2$. The commuting translations $\hat{S}_1 = e^{i\ell_Q\hat{P}_2/\hbar}$ and $\hat{S}_2 = e^{i\ell_P\hat{Q}_2/\hbar}$, $\ell_Q\ell_P = 2\pi\hbar$, stabilize the ideal grid state (Gottesman, Kitaev and Preskill, PRA **64**, 012310, 2001, metadata). With the aspect $\ell_P/\ell_Q = \alpha/\beta$, the Baker–Campbell–Hausdorff factor -1 gives $\hat{S}_1\hat{S}_2 = -e^{ic\hat{S}}$ with $c = \ell_P/(\hbar\alpha)$, and $2\pi/c = \sqrt{2\pi\hbar\alpha\beta} = \sigma$. Along the family, $\langle\hat{S}_1\rangle \rightarrow 1$ and $\langle\hat{S}_2\rangle \rightarrow 1$ mean $\|\hat{S}_k\psi - \psi\| \rightarrow 0$ for the unitaries $\hat{S}_k$, hence $\|\hat{S}_1\hat{S}_2\psi - \psi\| \rightarrow 0$ and $\langle e^{ic\hat{S}} \rangle \rightarrow -1$. So the law of $c\hat{S}$ modulo 2π concentrates at π , and under the force the law of $c(\hat{S} + d)$ concentrates at $\pi + cd$. When $d \notin \sigma\mathbb{Z}$ the two limit points differ, and a test on cT modulo 2π has error tending to zero. The momentum distribution's envelope has width of order $\hbar/w$, so $\Delta_2 = \lambda_2\Delta\hat{P}_2 \rightarrow \infty$. $\square$

At fixed teeth the error is small only when the signal's distance from $\sigma\mathbb{Z}$ exceeds the teeth's width, so “at any force” is a limit along the family, bought with diverging recoil; the grid probe stays above Theorem R.

Such states measure both quadratures of a small displacement at once, which is the principle of the grid-state displacement sensor of Duivenvoorden, Terhal and Weigand (PRA **95**, 012305, 2017, metadata). Its blind spots are the shifts in $\sigma\mathbb{Z}$, the translations whose Weyl phase with the stabilizers is a multiple of 2π : the grid reads the signal modulo a central phase, the structure the polygon-lift note isolates as the part no preparation can move. That the two phases are one quantity is a correspondence of structure, not derived here.

So Theorems B and C of the mark-cost note hold for Gaussian probes, where the Wigner function is a density and the record is the classical Gaussian model, and the correlation coefficient is the resource that lowers the floor within that class. Outside it the second moments do not bound the error, and Theorem R is the statement that holds.

4. Tightness

In the correlated protocol of Proposition D the outer marks are sharp, so their recoils are unbounded and the minimum in Theorem R takes the line through the endpoints, $c_1 = c_3 = 0$, $|c_2| = F\tau^2/8m$. The balanced middle mark has $2\delta_2 = \tau\Delta_2/2m$ and, for a pure Gaussian probe, $\delta_2\Delta_2 = \hbar/(2\sqrt{1-\rho^2})$, so $\Delta_2^2 = 2m\hbar/(\tau\sqrt{1-\rho^2})$. At its decision threshold $\tau\Delta E = 32z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$,

$$\Delta_2|c_2| = z_{1-\epsilon}\hbar\sqrt{\frac{2}{1+\rho}},$$

between $z\hbar$ and $\sqrt{2}z\hbar$ for every $\rho \in [0, 1)$. The balance $2\delta_2 = \tau\Delta_2/2m$ is the one that is optimal at $\rho = 0$, and at fixed ρ it wastes the factor $\sqrt{2/(1+\rho)}$. The best probe at a fixed recoil does better. The statistic's noise is $-(\alpha\hat{Q}_2 + \beta\hat{P}_2)$ with $\alpha = 2/\lambda_2$, and $[\alpha\hat{Q}_2 + \beta\hat{P}_2, \hat{P}_2] = i\hbar\alpha$, so Robertson gives $\Delta(\alpha\hat{Q}_2 + \beta\hat{P}_2) \geq \hbar\alpha/(2\Delta\hat{P}_2) = \hbar/\Delta_2$, attained by a pure Gaussian state of minimum uncertainty for that pair. The protocol then decides once $s\Delta_2 \geq 4z_{1-\epsilon}\hbar$, that is $\Delta_2|c_2| = z\hbar$, for every recoil. Theorem R asks for $\hbar\arcsin(1-2\epsilon)$, so the best Gaussian three-mark protocol sits within the factor $z_{1-\epsilon}/\arcsin(1-2\epsilon)$ of the bound, about 1.47 at five per

cent error, and uncorrelated probes within $\sqrt{2}$ times that. The bound in question is Theorem R with the line through the end marks; its constant-force form (4), with the line of best uniform approximation, charges the end marks too, and their unbounded recoil makes its left side infinite. The factor grows without bound as $\epsilon \rightarrow 0$, since $z_{1-\epsilon}$ does while $\arcsin(1 - 2\epsilon)$ stays below $\pi/2$; whether some non-Gaussian protocol meets the bound up to a constant at every ϵ is open here. The recoil-weighted sagitta is the right universal resource: squeezing moves the protocol along it and never below it.

5. What this changes

The three floors, ordered. Resource-free: the phase $\mathcal{K}_\tau/\hbar$ between the motion and its inscribed polygon (polygon-lift note, Theorems 1 and 7). Universal over probe states: the recoil bound $s \sum_j \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon)$ (Theorem R). Universal over instruments with a bounded laboratory: the aperture bound $F\tau L + \frac{F\tau^2}{2m}P \geq \hbar \arcsin(1 - 2\epsilon)$ (probabilistic note, with the kick form $\int |f|L \geq \hbar \arcsin(1 - 2\epsilon)$ of the path-length note). And for Gaussian probes with correlation at most ρ , the sharp floor $\tau\Delta E \geq 24z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$ (mark-cost note).

M3, restated. The mark-floor note's premise has two clauses, a product of resolution and recoil and the independence of the two. Theorem R shows that the recoil clause alone is universal: a mark that records the sagitta must leave the delivered impulse undetermined, with $s \sum \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon)$. In Newton's words from the mark-floor note, the premise that carries $h > 0$ for every probe is that a record costs an undetermined impulse, and the pairing is with the sagitta of Lemma X.

Beyond the momentum-transfer class. The proof uses only that the pointer's error is translated by the unitary generated by the delivered impulse, $[\hat{N}_j, \hat{D}_j] = -i\hbar$, and that the apparatus enters the record through $\hat{N}_j$ and $\hat{D}_j$. The commutation of the pointer with the body's momentum after any coupling gives $[\hat{N}, \hat{D}] = -i\hbar - [\hat{y}, \hat{D}] - [\hat{N}, \hat{p}]$, so Theorem R extends to every instrument whose error and recoil do not depend on the body (Arthurs and Goodman, PRL **60**, 2447, 1988, metadata). That is the next step, and the body-dependent case, where Ozawa's relation (PRA **67**, 042105, 2003, metadata) adds the body's own spreads, is the one after.

6. Consequence for STATE

Theorem A's extension, option 3 of the 2026-09-22 plan, is settled constructively. The Gaussian floor holds for Gaussian probes and grid probes break it (Proposition G); the universal statement over probe states is Theorem R, sagitta times total recoil at least $8\hbar \arcsin(1 - 2\epsilon)$, attained within a constant by squeezed probes at every squeezing. Next: option 1 (additive-noise instruments), then option 2 (Ozawa's relation).